NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science & Engineering
MID SEMESTER EXAMINATION, Oct. 2024

B. GTech. (UG) **Semester- 05**
Course Name: Database Management System **Code: CS34109**
Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. *Attempt All questions.*
2. *You may answer the questions in any order. However, all parts of the same question must be answered together.*
3. *Assume any suitable data, if necessary. Clearly state any reasonable assumption you make.*
4. *The Marks, CO (Course Outcome), and BL (Bloom's Level) related to questions are on the right-hand margin.*

		Marks	CO	BL
1.	A university wants to effectively manage its data, ensuring data integrity, security, and flexibility while meeting the needs of various stakeholders like end users, software developers (frontend and backend, and database administrator), and the system administrators who are involved in managing the database server. Most important requirement is if any changes are made in the lower level that will not affect the design and performance of the upper level, that is all the users will be independent of each other. Suggest proper architecture for the database design. Justify your answer with a diagram.	10	CO 1	A
2.	A small public library wants to create a computerized system to manage its books, patrons, and borrowing activities. The goals are to maintain a record of each book, including title, author, ISBN, publication year, and availability status; Store information about patrons, such as name, address, contact details, and library card number; and keep a log of books borrowed and returned by patrons, including due dates and fines. A book can have many authors and an author can write many books.The library also has a policy that a book can be borrowed by a patron, and a patron can borrow a maximum of 3 books at a time. (i) Identify all the entities and their attributes, mark attribute types. (ii) Identify all the relationships and their type and degree. (iii) Carefully design the ER diagram to create a robust and efficient system to manage its resources and services.	2+3+5	CO2	C

3.	A university assigns roll_numbers as a unique key to every newly admitted student, however each student carries many unique identities by which they can be uniquely identified. (i) Explain why the university assigned a new unique key? (ii) Identify the type of key "roll_number" is? (iii) Consider STUDENT(roll_number, first name, last name, address, date of birth, gender, emailID, phone), identify best possible superkey for STUDENT. List down all possible candidate keys for STUDENT.	3+2+5	CO 3,	E

****End of Questions****

Learning Outcomes:

CO 1. define data independence, data models, schema, and database instances. Also, distinguish among different database models and their practical implications.
CO 2. translate queries to Relational Algebra, Relational calculus and using Structure Query Language statements used in the creation and manipulation of database
CO 3. identify the methodology of conceptual modelling through Entity Relationship
CO 4. analyze and design an actual database application and develop and evaluate an actual database application
CO 5. Describe basic database storage structures, access techniques, and transaction processing.

Level	Bloom's Taxonomy
1	Remembering (R)
2	Understanding (U)
3	Applying (P)
4	Analyzing (A)
5	Evaluate (E)
6	Create (C)